MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : ES-CS401/PCC-CS 404/PCC-CS404/PCC-CSD 402/PCCCS404 Design & Analysis of Algorithms

UPID : 004416

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) What is the objective of the knapsack problem?

(II) Which traversal algorithm can be used to most efficiently determine the presence of a cycle in a given graph?

(III) Is it possible to select pivot element at random in quick sort?

(IV) A randomized algorithm uses random bits as input in order to achieve a _____ case good performance over all possible choice of random bits.

(V) What will be the recurrence relation of the following code?

```
int sum(int n)
{
  If(n==1)
    return 1;
  else
    return n+sum(n-1);
}
```

(VI) Define NP.

(VII) Every graph has only one minimum spanning tree. State True or False.

(VIII) Quick Sort is not a stable sorting algorithm in its typical implementation -True or False?

(IX) What is Cook's theorem?

(X) We can solve any recurrence by using Master's theorem - State true or false.

(XI) In what manner is a state-space tree for a backtracking algorithm constructed?

(XII) Which algorithm is used to find the shortest path between all pairs of vertices in a graph?

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Analyze the time complexity of merge sort algorithm using divide and conquer. Derive its recurrence relation before analysis. [5]
3. Explain graph coloring problem using backtracking. [5]
4. Explain randomized quick sort algorithm. [5]
5. Discuss diagrammatically the relations among P class, NP class, NP hard and NP complete. [5]
6. Analyze the time complexity of a single source shortest path algorithm which allows negative edge. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Explain the travelling salesman problem with one example. What is the time complexity of this problem? [5]

(b) Explain job sequencing with deadline algorithm using greedy strategy. [5]

(c) Using greedy method, find an optimal solution to the problem of job sequencing with deadline where n=4, (p1,p2,p3,p4)=(100,10,5,27) and (d1,d2,d3,d4)=(2,1,2,1), where n is total number of jobs, p[ ] and d[ ] are profits and deadlines respectively. [5]

8. (a) Define spanning tree of a graph. What is MST? [5]

(b) Explain Kruskal's Algorithm for finding out MST of a ~~given~~ graph. [5]

(c) Analyze the complexity of Prim's algorithm. [5]

9. (a) Apply recursive substitution method to solve the recurrence $T(n) = 2T(n-1) + 1$ with initial condition $T(0) = 0$ [5]

(b) Apply Master's Theorem to solve $T(n) = 4T(n/2) + n^2$ [5]

(c) For the given set of 5 items and the knapsack capacity of 10 kg, find the maximum profit applying fractional knapsack problem using greedy method. Weights and Profits are given below: [5]

Weights, Wi = {3, 3, 2, 5, 1}

Profits, Pi = {10, 15, 10, 12, 8}

10. (a) Prove that Clique Decision Problem in NP-C. [5]

(b) Explain a graph traversal algorithm which uses backtracking. [5]

(c) Explain Transitive Closure. [5]

11. (a) What is an approximation algorithm? [3]

(b) What are the features of an approximation algorithm? [5]

(c) Describe some examples of approximation algorithms. [4]

(d) What do you mean by performance ratios of the approximation algorithm? [3]

*** END OF PAPER ***

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : MC 401/MC-401/MC401 Environmental Sciences

UPID : 004421

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(i) What are the differences between Species, Population and Ecological Community ?
(ii) What are the differences between Pollutants and Contaminants ?
(iii) What are the main river water pollutants ? What is the range of Dissolved Oxygen concentration for healthy water ?
(iv) What is Lithosphere and what are its major components ?
(v) Which Smog is called "London Smog" and "Los Angeles Smog" ?
(vi) How many main nutrients do we have in our environment ? Name them.
(vii) What is a steady state system ? Indicate by means of mathematical term.
(viii) What are the differences between Food Chain and Food Web ?
(ix) What is Ozone layer depletion ? What is the full form of CFC and what are sources of CFCs ?
(x) What are the full form of BOD and DO ? What are the most common heavy metals found in water ?
(xi) What are the sources of Solid Waste ? What do you understand by 'Resource Recovery' from solid waste ?
(xii) What is an aquifer ? How is water available in an aquifer ?

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. ~~(i) What is Environmental Laws in India ?~~ [5]
(ii) What are the differences between Law and Act ?

3. ~~(i) What is environment and what are its functions ?~~ [5]
(ii) What is the relationship among Man, Society and Environment ? Show by flow diagram.
~~(iii) What do you understand by sustainable development ?~~

4. (i) What do you understand by 'Biogeochemical Cycle' ? How is it important to the environment ? [5]
~~(ii) How many main nutrient cycle are available for maintaining balance of environment ? Name them.~~

5. ~~(i) What is Smog and what are two distinct types of Smog ?~~ [5]
(ii) What are the differences between the two different types of Smogs ?
~~(iii) What are CFCs and what are their sources ?~~

6. ~~(i)~~ How resource is recovered from solid waste ? [5]
~~(ii) What is land filling of solid waste ?~~
~~(iii) What is composting ?~~

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (i) What do you understand by Air Quality standard ? 2 [15]
(ii) How many types of air quality standards do we have? Name them. 5
(iii) Why do we need air quality standards ? 5
(iv) What do you understand by ambient air quality standards ? 3

8. (i) What are water hardness, alkalinity and water softening ? 6 [15]
(ii) What are toxic pollutants in water and what are their sources ? 5
(iii) Is Arsenic a toxic pollutant and how Arsenic is found in groundwater ? 4

9. (i) What are the effects of Suspended Particulate Matters ? [4 + 4 +3+ 4]
(ii) What are effects of oxides of Nitrogen ?
(iii) Which oxide of Nitrogen is called laughing gas ?
(iv) What are the two important oxides of Sulphur ?

10. (i) What is Noise Pollution ? [3 + 3 +4 +5]
(ii) What are the main sources of Noise Pollution ?
(iii) What is the formula to calculate sound pressure ?
(iv) Determine the sound pressure level, if the rms pressure value is 30 μPa .

11. (i) What is Environmental Impact Assessment ? [3 + 4 +5+3]
(ii) What are various steps of EIA process ?
(iii) What is Environmental Auditing and its purposes ?
(iv) Name five important Environmental Acts of India.

*** END OF PAPER ***

MAKAUT 1059

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : BS-B401/BS-BIO401/BSC 401/BSC-401/BSC401 Biology

UPID : 004408

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) The genotypic ratio of monohybrid experiment is _____.

(II) _____ virus infects bacterial cells.

(III) Who is called the "Father of Taxonomy"?

(IV) Thymine is a ____ base.

(V) _____are the esters of fatty acid and glycerol.

(VI) The position of a gene on a chromosome is called _____.

(VII) What is shine dalgarno sequence?

(VIII) _____ bacteria are used for curd formation.

(IX) Product of Krebs cycle essential for oxidative phosphorylation is _____.

(X) H2 donor during photosynthesis is _____.

(XI) During the process of Gram staining, Gram-Negative bacteria appears _____ in colour.

(XII) Write one example of ureotelic organism.

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. What is the induced fit theory of enzyme? [5]
3. Explain the structure of nucleosome with diagram. [5]
4. Explain testcross. Write it's significance. [5]
5. Differentiate between unicellular and multicellular organisms with example. [5]
6. What is gluconeogenesis? Differentiate between Nucleotides and Nucleosides. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. What are Model organisms? Give characteristics of of *E.coli, A. Thaliana, M. musculus* as model organisms. [15]
8. (a) What are the three Laws of Inheritance proposed by Mendel? [10]
   (b) Explain Dihybrid cross. [5]
9. (a) Define enzyme inhibition. Explain the different types of inhibitions with suitable diagram. [10]
   (b) Explain the factors affecting enzyme activity. [5]
10. (a) Write the name and functions of main three different types of RNA. [6]
    (b) What are phospholipid and glycolipids? [4]
    (c) Differentiate between saturated and unsaturated fats. [5]
11. (a) Differentiate between facilitated diffusion and active transport with diagram. [10]
    (b) Explain the tertiary structure of protein. [5]

*** END OF PAPER ***

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PCC-CS 403/PCC-CS403/PCC-CSB$401/PCCCS403 Formal Language & Automata Theory

UPID : 004423

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(i) What type of languages can a Pushdown Automaton accept?
(ii) What is the "halting state" in a Turing Machine?
(iii) What does it mean for a problem to be undecidable?
(iv) What does an edge in a transition diagram represent?
(v) What type of languages can regular grammar describe?
(vi) What kind of automaton is used to parse a Context-Free Language?
(vii) What does a PDA do if the stack is empty during a transition?
(viii) Can a Turing Machine simulate a finite automaton?
(ix) Why is state minimization necessary in DFA?
(x) What is the significance of an initial state in a transition diagram?
(xi) Write the regular expression for the set of all strings of 0's and 1's beginning with 0 and ending with 1.
(xii) When is a grammar called type-2 or context-free grammar?

Group-B (Short Answer Type Question)

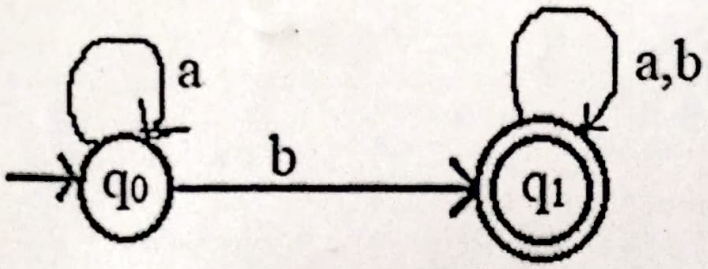
Answer *any three* of the following : [5 x 3 = 15]

2. Prove (1+00*1) + (1+00*1) (0+10*1)* (0+10*1) = 0*1 (0+10*1)* [5]
3. If G is the grammar with P = {S→S+S | S*S | a}, show that G is ambiguous. [5]
4. Design a pda P accepting the set of all strings over {a, b} with an equal number of a's and b's. [5]
5. Draw the state Transition Diagram of Turing machine for 2's complement of any binary string. [5]
6. Derive the transition diagram for a DFA that accepts strings over {a,b} that have exactly two a's and any number of b's. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Find a reduced grammar of P={S→AB|CA, B→BC|AB, A→a, C→aB|b}. [6]
   (b) Convert this grammar into Chomsky normal form. P={S→aAD, A→aB|bAB, B→b, D→d} [5]
   (c) Remove NULL productions from this grammar. P={S→aS|AB, A → ^, B → ^, D → b} [4]
8. (a) What are the differences between deterministic pushdown automata and non-deterministic pushdown automata? [3]
   (b) Design a PDA for the Context Free Language L= $\{a^n b^n \mid n>=1\}$. [6]
   (c) Construct a pda A equivalent to the following CFG. P= {S→0BB, B →0S | 1S | 0} [6]
9. (a) What is the concept of the tape in a Turing Machine, and how does it function? [4]
   (b) Draw the state Transition Diagram of Turing machine for 1's complement of any binary string. [4]
   (c) Design a Turing Machine to recognize all strings consisting of even numbers of 1's. [7]
10. (a) What is a Universal Turing Machine (UTM)? [2]
    (b) What is the significance of the Halting Problem in computational theory? [3]
    (c) Describe how UTM simulates the behavior of any other Turing machine. [5]
    (d) What are some common undecidable problems related to formal languages and Turing machines? [5]
11. (a) Write the language L for the grammar G=({S,A}, {a, b, c}, P, S) and P={S→aS, S→A, A→bAc|bc}. [3]
    (b) Construct a grammar generating L = $\{a^n b^n c^i \mid n>=1, i>=0\}$ [4]
    (c) Find the Regular Grammar from the DFA given below. [8]



(d) Show that L={a^p | p is a prime} is not regular. [5]

*** END OF PAPER ***

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PCC-CS402 Computer Architecture

UPID : 004442

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) What is the function of the Instruction Register (IR)?
(II) What is the difference between SRAM and DRAM?
(III) What is the role of the MAR (Memory Address Register)?
(IV) What is the difference between RISC and CISC architectures?
(V) What is the purpose of a Bus in computer architecture?
(VI) What is the function of a Decoder in a CPU?
(VII) Explain the difference between I/O-mapped I/O and memory-mapped I/O.
(VIII) What is the purpose of an interrupt vector table?
(IX) What is instruction-level parallelism (ILP)?
(X) What is the role of a shadow register in computer architecture?
(XI) What does the term 'speculative execution' refer to in modern processors?
(XII) Which type of parallelism does SIMD (Single Instruction, Multiple Data) exploit?

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. What are the main differences between primary, secondary, and cache memory? [5]
3. Define pipelining and explain its stages. [5]
4. Differentiate between hardwired control and microprogrammed control. [5]
5. Describe the working of an arithmetic logic unit (ALU) with a diagram. [5]
6. Explain the concept of Register Renaming and how it mitigates Write-After-Write (WAW) and Write-After-Read (WAR) hazards in modern processors. Provide a practical example. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Explain the concept of virtual memory and its advantages. [8]
   (b) What are the different types of addressing modes in computer architecture? Provide examples. [7]
8. (a) Explain the concept of instruction cycle and its stages. [8]
   (b) Describe the function of the control unit in a CPU. [7]
9. (a) Discuss the role of cache coherence in multiprocessor systems. [5]
   (b) Describe the key functions of device controllers in computer systems. [10]
10. (a) Discuss the importance of performance metrics such as MIPS and FLOPS. [5]
    (b) Explain the role of DMA (Direct Memory Access) in improving system [10]
11. (a) Describe the architecture and working of RAID (Redundant Array of Independent Disks) levels. Discuss the pros and cons of different RAID levels. [8]
    (b) Explain the concept of Data Forwarding in pipelined processors and how it resolves data hazards. Provide a practical example. [7]

*** END OF PAPER ***

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PCC- CS401/PCC-CS401/PCCCS 401/PCCCS401 Discrete Mathematics

UPID : 004407

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) Find the domain of $f(x)$ where $f(x)=\sqrt{\log\frac{4x-x^2}{3}}$, $x, f(x)$ are real.

(II) What is the rank of the word "SACHIN" in dictionary wise?

(III) Write the CNF form of $(p \rightarrow (q \wedge \neg r))$.

(IV) State Lagrange's theorem on group.

(V) Find the minimum and maximum number of edges of a simple graph with 10 vertices and 3 components.

(VI) If $aN=\{ax: x \in N\}$ then $7N \cap 3N =$

(VII) How many ways are there to travel in x-y-z space from the origin (0,0,0) to the point (4,3,5) by taking unit steps in positive x,y,z directions only?

(VIII) Express the proposition $p \leftrightarrow q$ in the minimal complete set of connectives $\{\neg, \vee\}$.

(IX) Does the ring Z_6 of integers modulo 6 is integral domain? Justify.

(X) Define separable graph.

(XI) Define enumerable set and give an example.

(XII) How many square are there in chess board?

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Show that a cycle with n vertices is a perfect graph if n is even. [5]

3. Let D_{36} be the set of all positive divisors of 36. Show that D_{36} is a POSET w.r.t.the relation 'divisor'. Draw the Hasse diagram of this POSET. Find the Supremum and Infimum of the set {4, 9}. [5]

4. How many eight bit binary strings are possible that never begin either with 011 or with 101? [5]

5. Check the validity of the following argument : [5]
"If my program runs successfully, then I will submit my project. I can appear in the examination only if I submit my project. Either my program runs successfully or the computer crashes. Therefore, if the computer does not crash then I can appear at the examination".

6. Find an equivalent expression for $p \wedge (q \leftrightarrow r) \vee (r \leftrightarrow p)$ that contains minimal complete set of connectives. [5]

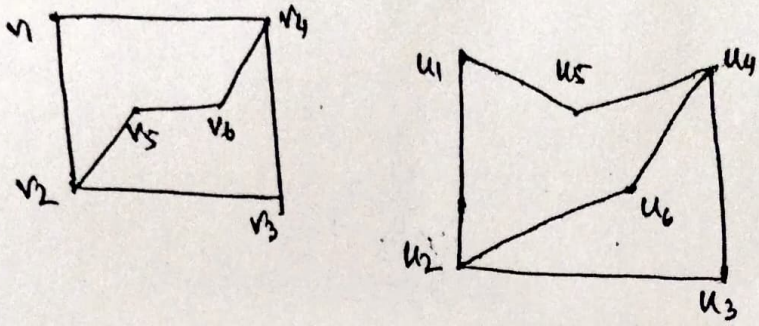
Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Show that a simple graph with n number of vertices and k number of components can have at most (n-k)(n-k+1)/2 edges. [5]

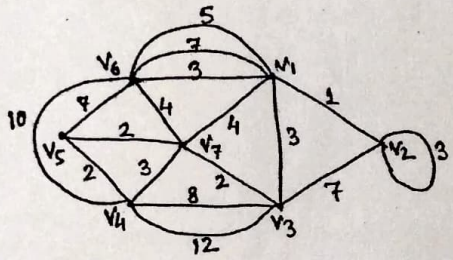
(b) Show the following: [5]

Show that the following two graphs are isomorphic:-



(c) Solve the problem: [5]

By Dijkstra's Algorithm find the shortest path and shortest distance from vertex v_2 to vertex v_5 in the following graph.



8. (a) Find the total number of integers between 1 and 1000 which are neither perfect square nor perfect cubes? [5]

(b) Find the number of ways a company can assign 7 projects to 4 people so that each person gets at least one project. [5]

(c) Consider 6 people. Any two of them are either friends or strangers. Find the number of selected persons where the persons are either mutual friends or mutual strangers. Explain the method. [5]

9. (a) Find all solutions in positive integers of $5x+3y=52$. [5]

(b) If $\gcd(a,b)=1$, prove that $\gcd(a+b, a^2-ab+b^2)=1$ or 3. [5]

(c) Show that the product of any r consecutive natural numbers is divisible by r! [5]

10. (a) In an College offering a course in which every student will have to take English and one of the following groups (Botany, Zoology, Microbiology), (Botany, Chemistry, Zoology), (Physics, Zoology, Botany), (Maths, Physics, Chemistry), (Maths, Physics,Statistics),(Maths, Statistics, Computer), (Computer, Chemistry, Microbiology). Find the period allocation in a day such that students can attend their classes without any hindrance. [5]

(b) Show that s is a valid conclusion from the premises: $p \to \neg q, q \vee r, \neg s \to p, \neg r$ [5]

(c) Write the converse, inverse and contrapositive of $(p \wedge q) \to \neg q$. [5]

11. (a) Also write the contrapositive statement of the statement: "If a triangle is right angled then it is not equilateral" [5]

Let GL(2, R) be the multiplicative group of all real non singular matrices of order 2. Show that the set

$H = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : ad - bc = 1 \right\}$ is a subgroup of $GL(2, R)$

(b) Show that the set G of all ordered pairs (a, b) with $a \neq 0$, of real numbers a, b forms a group with operation '$\circ$' defined by $(a, b) \circ (c, d) = (ac, bc + d)$ [5

(c) Let G be an abelian group. Prove that the subset $S = \{p \in G : p = p^{-1}\}$ forms a subgroup of G. [5

*** END OF PAPER ***

MAKAUT 1059